How accurate LIGHT_Kr v0_2 is

The pure-krypton model LIGHT_Kr v0_2, version 0.2.0-kr, on 14 October 2026: LIGHT v0_5's krypton network without its fluorine, its electron collisions resolved level by level and Hornbeck–Molnar ionization as a band, scored against pure-krypton measurements, with its propagated uncertainty, its reference point and its checks

Bottom line. No measurement in pure krypton covers v0_2's own regime, a gas at 0.1 to 2 atm whose electrons a field heats. The measurements that exist test its parts, and none of them set a rate.

- **The electron kinetics.** The two-term Boltzmann solver on Biagi's level-resolved cross sections reproduces krypton's measured drift velocity [34, 35] to 10.8 % rms over 41 points, and the ionization coefficient [36, 37] to 7.6 % rms from 100 to 300 Td. Below 50 Td its mean error is -47 % (v0_1's was -54 %). That error is not the solver's: a Monte Carlo solution of the same cross sections [3] agrees with it to 4.0 % at 20 to 50 Td. Every published set of krypton cross sections falls below these measurements, which Bordage et al. attribute to Penning ionization of impurities in the experiments [2], and the two measurements themselves differ by up to $\times 2.2$. Pure krypton has one more process there, the associative (Hornbeck–Molnar) ionization of atoms excited above 12.85 eV [4]. Its share is not measured, so v0_2 carries it as a band at 0. At the band's top 18 of the 21 points below 50 Td lie between its two ends, but the same top makes the model's mean error from 100 to 300 Td +39 %, so a share that large cannot hold at every field (Section 4).
- **Excitation to light, under an electron beam.** At low pump v0_2 turns 0.550 of the deposited energy into 147-nm light. Morozov et al. measured 0.42 ± 0.05 [39], so the error is +31 %. v0_2 makes one photon per excitation and per ion, which is what Saito et al.'s measurement implies (0.547 [38]). At 3×10^5 W cm^{-3} and 1 atm it gives 0.345, 37 % below its ceiling, because Penning ionization of Kr* costs 0.669 photons per ion pair. Eckstrom et al. report a value near their own limit up to that pump rate [40].
- **At LIGHT's reference point** (1 atm, 300 kW cm^{-3} for 1 μs) v0_2 sends 14.20 % of the input to 147 nm on the kinetic EEDF. Propagated from its sources, that share carries ± 11.9 points in quadrature (± 84 %). The largest term, 75 % of the variance, is the excimer's formation, Kr* + 2Kr, whose measurements spread by a factor of 3.2.
- **What decides the answer** is the electrons' 5s $\leftrightarrow$ 5p cycle on Jung et al.'s measured cross sections [10]. The electrons pump Kr* to the 5p rung 34 times for each Kr_2^* formed, and krypton's relaxation of the rung turns 77.6 % of the input into heat.
- **The numerical error is far smaller.** The energy ledger closes to 3×10^{-15} of the input. Halving the kinetic closure's renewal steps or doubling its energy cells moves the 147-nm and near-IR shares by at most 3×10^{-5} of themselves.

Table 1 lists every test, Section 1 reads them as errors and propagates the model's own uncertainty, and the sections after it give each test's detail. No measurement in the table was used to set a rate: the model's rates come from the sources in Appendix A, and the data here only score them.

test	measured	v0_2	error	role
energy ledger closure, reference point		3×10^{-15} of E_{dep}		bookkeeping
energy balance at every node of the flow network		20 nodes, to 10^{-8}		bookkeeping
drift velocity, Pack et al. 1962 [34, 35], 41 points, 0.0008 to 3.06 Td			rms 10.8 %	solver check
ionization coefficient, Kruithof 1940 and Heylen 1971 [36, 37, 35], 100 to 300 Td			rms 7.6 %	solver check
the same, 14 to 50 Td		direct ionization; the Hornbeck–Molnar band at its top	mean −47 %; +87 %	solver check
ionization coefficient, Monte Carlo on the same cross sections [3], 20, 30, 40 and 50 Td			+3.7, −4.0, −3.5, −2.5 %	solver check
energy per excitation or ion under a beam, Saito et al. 2003 [38]	15.4 eV	15.38 eV	−0.1 %	the beam's check
VUV efficiency, continuous e-beam, Morozov et al. 2008 [39]	0.42 ± 0.05	0.550	+31 %	held out
VUV efficiency at 3×10^5 W cm^{-3}, Eckstrom et al. 1988 [40]	near 0.46, their limit	0.345 at 1 atm, 37 % below its ceiling; 0.529 at 2 atm, 4 % below	−25 % at 1 atm	held out

Table 1: LIGHT_Kr v0_2 against the data, every band at its default, on the kinetic EEDF. An error is the model less the measurement over the measurement, signed for a single value and rms (or a mean) over a set; in the Monte Carlo row, v0_2's two-term solution less the Monte Carlo's over the Monte Carlo's at each E/N. The beam rows run under the e-beam drive of Section 5.

1 Overall accuracy and its uncertainty

Against the data. Three groups of measurements bear on v0_2, and each tests a part of it (Table 1).

- **The electron swarm** (Section 4) scores the solver that gives every electron rate. The drift velocity is within 10.8 % rms. The ionization coefficient is within 7.6 % from 100 to 300 Td, −14 % on average from 50 to 100 Td and −47 % below 50 Td, where every published cross-section set falls below the measurements and the two measurements differ from each other by up to ×2.2. A Monte Carlo solution of the same cross sections agrees with the solver to 4.0 %.
- **The e-beam measurements** (Section 5) score the chain from excitation to the 147-nm photon. At low pump the error against Morozov et al. is +31 %, 2.6 of their standard errors. At 3×10^5 W cm^{-3} and 1 atm it is −25 % against Eckstrom et al.'s report, and it loses 37 % below its own ceiling where their beam lost little.
- **Nothing scores the field-driven cycle.** No pure-krypton measurement in hand tests the 5s ↔ 5p cycle under a field, which decides v0_2's answer. Its cross sections are measured, and their error is propagated below.

Propagated from the sources. Each input is set to the two ends its source allows and the run repeated, the others at their defaults, on both electron closures (Table 2). The inputs are the 8 bands the rows read and the 3 measured constants with a quoted error that no band covers: $Kr_2^+ + e^-$ (±10 % [30]), $Kr^+ + 2Kr$ (the spread of the three measurements [26, 27, 28]) and the Kr_2^* lifetime (270 ± 5 ns [25]). Half the difference between an input's two ends is its term. The quadrature sum treats the terms as independent standard deviations, which a band's range is not, so it is indicative. The excursions from the reference, summed by sign, are the ceiling of the spread, not a bound. Biagi's cross sections are not propagated; Section 4 scores them. Nor is the 5p rung's lifetime, whose light is under 1 % of the input.

On the 147-nm share at the reference point, 14.20 % of the input on the kinetic EEDF, the quadrature sum is ±11.9 points (±84 %). The excursions sum to −16.7 and +31.6 points. The largest terms are

Kr^* + 2Kr (10.36 points), Kr_2^* + Kr_2^* (3.23) and the 5s → 5p cross section (3.21). On the two-temperature closure the sum is ±12.2 points (±76 %).

The formation term dominates because Kr_2^* forms in competition with the electrons' pumping of Kr^* to the 5p rung, so a faster formation lets more Kr^* escape the cycle. The default, 1.7×10^{-32} $cm^6 s^{-1}$, is the one measurement made at the pressures the model runs at, 0.5 to 1 atm in an e-beam experiment [19]. The other published values, 1.8 to 4.4×10^{-32}, come from low-pressure afterglows [20, 21, 22], and the band's top is Hughes's 5.5×10^{-32} as Quigley and Hughes print it [23]. Four inputs with no measurement at all are 0 by default. Three of them, Kr_2^* + Kr_2^* and the two stepwise ionizations, each at its band's top cut the share by a fifth to a half. The fourth, the Hornbeck–Molnar share, at its top moves it by −0.02 %: direct and associative ionization make a small part of the ions next to Penning's.

input	its two ends	147 nm, kinetic	147 nm, two-temperature	near IR, kinetic	T_g (K), kinetic
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} to 5.5×10^{-32} $cm^6 s^{-1}$	14.20 to 34.92	16.00 to 37.08	0.64 to 0.45	807 to 686
Kr_2^* + Kr_2^*, Penning	0 to 5.0×10^{-10} $cm^3 s^{-1}$	14.20 to 7.73	16.00 to 8.55	0.64 to 0.69	807 to 846
$e^- + Kr^* \rightarrow Kr^{**}$, scale	0.5 to 1.4	18.76 to 12.35	21.12 to 13.86	0.58 to 0.67	780 to 818
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0 to 1.5×10^{-7} $cm^3 s^{-1}$	14.20 to 8.75	16.00 to 11.14	0.64 to 0.64	807 to 840
Kr^* + Kr^*, Penning	1.5×10^{-10} to 5.0×10^{-10} $cm^3 s^{-1}$	18.53 to 14.20	20.12 to 16.00	0.61 to 0.64	782 to 807
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0 to 2.0×10^{-7} $cm^3 s^{-1}$	14.20 to 11.56	16.00 to 14.32	0.64 to 0.65	807 to 823
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} $cm^3 s^{-1}$	15.74 to 14.20	17.54 to 16.00	3.41 to 0.64	782 to 807
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	±10 %	13.99 to 14.39	15.82 to 16.16	0.65 to 0.64	809 to 806
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.2×10^{-31} to 3.3×10^{-31} $cm^6 s^{-1}$	14.13 to 14.42	15.92 to 16.23	0.65 to 0.64	808 to 806
Kr_2^* lifetime	265 to 275 ns	14.19 to 14.21	15.98 to 16.01	0.64 to 0.64	807 to 807
$Kr(nl) + Kr \rightarrow Kr_2^+ + e^-$, the share	0 to 1	14.20 to 14.20	16.00 to 16.00	0.64 to 0.64	807 to 807

Table 2: The propagated uncertainty's inputs, each alone at its two ends with the others at their defaults, and the outputs at those ends (low end to high end), percent of E_{dep}. With every input at its default the 147-nm share is 14.20 % (kinetic) and 16.00 % (two-temperature).

Numerical accuracy.

- **The ledger.** The energy ledger closes to 3×10^{-15} of the input on the kinetic EEDF and to 8×10^{-16} on the two-temperature closure. The energy flows also balance at each of 20 nodes to 10^{-8}: the electron gas, each reservoir, each group of rows and each output.
- **The kinetic EEDF at the reference point.** The run takes 381 kinetic solutions, 277 in the pulse, at E/N from 1.32 to 15.5 Td. Over the pulse's second half the field carries 1.0001 to 1.0002 of the input. There the kinetic and the model's energy balances agree to 9×10^{-6} of the input, and the kinetic mean energy equals the model's to 3×10^{-11}.
- **Convergence.** Half the renewal step moves the 147-nm share by 1×10^{-5} of itself. A renewal at least every 5 ns moves it by 3×10^{-5}, and twice the energy cells by 3×10^{-5}. A ten times tighter solver tolerance moves it by 2×10^{-7}, the EEDF renewed 381 times against the reference's 381.
- **The change from v0_1.** On Biagi's four groups v0_1 gave the reference point's 147-nm share as 14.06 % (kinetic) and 15.05 % (two-temperature); v0_2 gives 14.20 and 16.00 %. The pulse's mean T_e moves from 2.29 to 2.24 eV on the kinetic EEDF.
- **The swarm scores** are independent of the grid: half and twice the 1000 energy cells move every rms by 0.18 points at most.

2 What v0_2 is

LIGHT_Kr v0_2 is a 0-D model: one uniform box of pure krypton at constant volume, into whose electrons a given power is put. It follows where that energy goes: into the 5s manifold (Kr^*) and the 5p rung (Kr^{**}), the ions Kr^+ and Kr_2^+, the excimer Kr_2^* and its 147-nm second continuum, the near-infrared 5p → 5s lines, heat, and the electron gas. Its energy ledger closes by construction. Every species carries a formation energy, and every row's energy goes to the electrons, the gas, light or storage.

Its network has 85 rows (Appendix A):

- electron-impact excitation of each of the 51 levels of Biagi's cross sections at its own threshold,

ionization and elastic loss [1, 6, 7]: the four 5s levels make Kr^*, every higher level the 5p rung Kr^{**} (the higher levels cascading through it), the excess over the product's energy heating the gas;
- the Hornbeck–Molnar ionization of the 20 levels at or above Kr_2^+'s energy [4], a band at 0;
- Penning ionization of Kr^* by Kr^*;
- ion conversion $Kr^+ \rightarrow Kr_2^+$, and dissociative recombination of Kr_2^+ onto the 5p rung;
- three-body excimer formation and its 147-nm emission;
- the rung's relaxation on krypton and its near-infrared radiation;
- the electrons' 5s $\leftrightarrow$ 5p cycle on Jung et al.'s measured cross sections [9, 10];
- and, as bands at 0, the rows with no measurement.

The electrons' rates come from the two-term Boltzmann equation with electron-electron collisions [8], solved during the run at the model's own mean energy (the default, minutes a run). The alternative is a Maxwellian with a tabulated tail (the two-temperature closure, seconds a run). A rate known only as a published spread, or not at all, is a band: a default and two ends.

What changed from v0_1. v0_1 read Biagi's older set, which groups krypton's excitations into four effective levels (5s, 5p, the 4d and 6s group, 5d and above) at the groups' lowest thresholds. v0_2 reads his level-resolved set (Magboltz 8.97), so every excitation costs its own level's energy, the electrons' distribution meets each threshold where it lies, and the species sit at the levels' measured energies: Kr^* at 9.9152 eV, Kr^{**} at 11.30 eV, Kr^+ at 13.9996 eV. The Hornbeck–Molnar rows are new. The rest is v0_1's: LIGHT v0_5's krypton network and machinery without its fluorine, and two entries of Jung et al.'s tables corrected to the paper [10]. At the reference point the 147-nm share moves from 14.06 % to 14.20 % on the kinetic EEDF (+1.0 %) and from 15.05 % to 16.00 % on the two-temperature closure (+6.3 %).

3 The reference point

LIGHT's drive in pure krypton: 1 atm at 300 K, 300 kW cm^{-3} for 1 μs into the electrons, a 10^{15} cm^{-3} seed, 2 μs in all. Table 3 gives its outputs on both closures.

	kinetic EEDF (default)	two-temperature
147 nm, Kr_2^*	14.20	16.00
near IR, 5p $\rightarrow$ 5s	0.64	0.63
heat, reactions and relaxation	83.11	81.40
heat, elastic collisions	2.58	2.49
stored in ions (the seed's ions recombine)	-0.74	-0.74
to Kr^* and Kr^{**} by electron impact, gross	21.31	24.56
to Kr^+	6.39	8.04
to Kr_2^* (formed $\times$ 8.43 eV)	14.38	16.20
Kr^* formed that the electrons pump to 5p (%)	95.7	94.8
Kr^* formed that becomes Kr_2^* (%)	2.8	3.2
5s $\rightarrow$ 5p pumpings per Kr_2^* formed	34	29
T_e, pulse mean (eV)	2.24	2.00
T_g at the end (K)	807	797
n_e at the end (cm^{-3})	4.4×10^{12}	4.6×10^{12}

Table 3: The reference point at t = 2 μs, percent of E_{dep} unless a row says otherwise. The fate rows (the first five, with the other stores) sum to 100; the flows below them overlap.

The electrons pump Kr^* to the 5p rung faster than three-body collisions turn it into Kr_2^*. Krypton relaxes the rung back, and the 1.388-eV gap of every cycle becomes heat: 77.6 % of the input, 46 eV for each 147-nm photon. On the kinetic EEDF the electrons sit at 2.24 eV, and the share differs from the two-temperature closure's by -11 %. The cycle's measured cross section therefore decides the answer, with Kannari's fit in its place as a comparison [11, 12]:

5s → 5p cross section	0 (no cycle)	× 0.5	× 1, Jung et al.	× 1.4	Kannari's fit
147 nm, kinetic EEDF	50.9	18.8	14.2	12.4	15.4
147 nm, two-temperature	52.2	21.1	16.0	13.9	17.5

Table 4: The 147-nm share (percent of E_{dep}) against the scale on the measured 5s → 5p cross section. The band runs from 0.5, the two metastable levels alone, to 1.4, the measurement's upper error.

Jung et al. measured the excitation out of the two metastable levels, $1s_5$ and $1s_3$, into the ten 5p levels, with stated uncertainties of ±35 and ±40 % [10]. v0_2's Kr* is the whole 5s manifold, so the two $J = 1$ levels, for which nothing is measured, take the set of the metastable with the same ion core. The lumped cross section is 1.1×10^{-14} cm^2 at 6 eV (5.5×10^{-15} from the metastables alone). Its Maxwellian rate is ×1.40 Kannari's fit at 1 eV, ×1.23 at 2 eV and ×1.20 at the reference point's 2.24 eV.

Pressure and power move the share as much (Table 5, two-temperature closure). It ranges from a fraction of a percent at 0.1 atm to 62 % at 2 atm and 100 kW cm^{-3}. At 0.1 atm a microsecond of this drive heats the gas to 2042 to 8750 K, beyond the measured range of several rates.

kW cm^{-3}	0.1 atm	0.25 atm	0.5 atm	1 atm	2 atm
100	0.2	1.6	8.7	29.2	61.9
300	0.1	0.6	3.4	16.0	46.1
500	0.0	0.3	2.1	11.0	37.7

Table 5: The 147-nm share (percent of E_{dep}) over pressure and power, 1 μs pulses, the two-temperature closure.

4 The electron swarm

The kinetic EEDF's solver is run as a swarm: no electron-electron collisions, the temporal growth term on, and every grid sized so that the EEDF falls 11 to 25 decades across it. The drift velocity is the Joule power over eE and the ionization coefficient $\alpha/N = k_i/W$. The data are the printed values of Dutton's survey [35], Figure 1. Biagi's cross sections were built to reproduce swarm data [1], so this is chiefly a check of the two-term solution and its numerics.

With the Hornbeck–Molnar band above 0, a share of the excitations of the 20 levels at or above Kr_2^+'s energy, 12.86 to 13.6 eV, ionizes associatively, Kr(nl) + Kr → Kr_2^+ + e$^-$ [4]. A steady-state Townsend experiment counts those electrons with the direct ones, so the swarm's coefficient is then $(k_i + k_{assoc})/W$. The band's share is a probability, 0 to 1; at its default of 0 the coefficient is direct ionization alone.

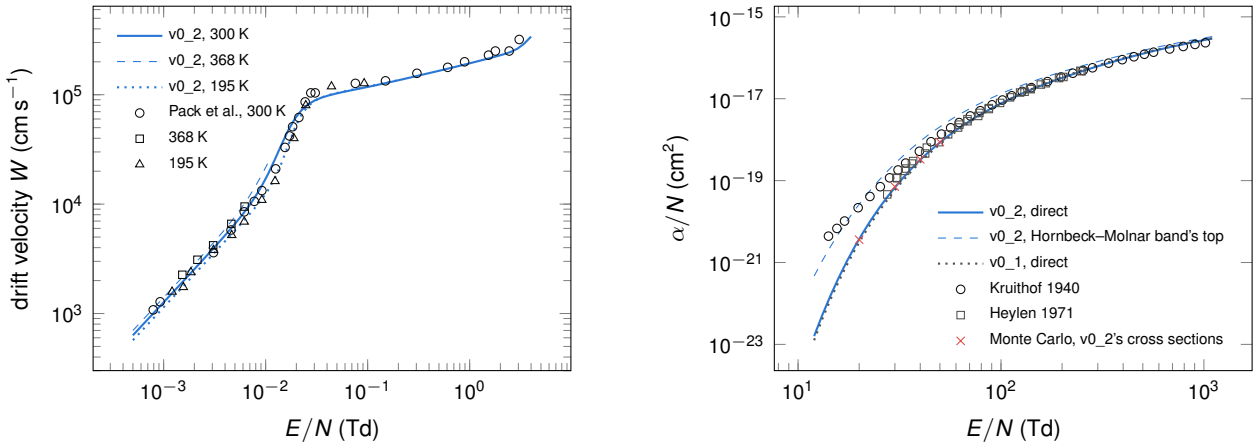


Figure 1: Pure krypton's drift velocity (left) and first Townsend coefficient (right): v0_2's two-term solution on Biagi's level-resolved cross sections against the measurements as Dutton's survey prints them [35]. Right: direct ionization (solid), the Hornbeck–Molnar band's top (dashed), v0_1 on Biagi's four groups (dotted) and a Monte Carlo solution of v0_2's cross sections [3] (crosses).

- **The drift velocity** of Pack, Voshall and Phelps [34], 41 points at 195, 300 and 368 K: rms 10.8 % (v0_1, 9.6 %), mean −0.5 %. It is 6.8 % rms below 0.01 Td, where the electrons are thermal. Over the runaway from the Ramsauer minimum, 0.01 to 0.05 Td, it is 16.2 %: there W rises 5-fold while E/N rises 2.4-fold. From 0.05 to 3 Td it is 8.5 % (mean −7.5 %). The largest deviation is +27 % at 0.0125 Td.
- **The ionization coefficient** of Kruithof [36] and Heylen [37], 66 points to 1017 Td (Table 6). It is 7.6 % rms from 100 to 300 Td and +9 % on average above. There the mean energy passes 7 eV, where the two-term approximation and the steady-state Townsend geometry part ways. From 50 to 100 Td it is −14 % on average. Below 50 Td it is −47 %: −68 % against Kruithof and −23 % against Heylen, whose values fall increasingly below Kruithof's as E/N decreases. At 14.1 Td the model is ×0.027 Kruithof's.

E/N (Td)	14 to 50	50 to 100	100 to 300	300 to 1017
points	21	16	20	9
v0_1, direct, mean	−54	−19	−4	+12
v0_2, direct, mean	−47	−14	−7	+9
against Kruithof, Heylen	−68, −23	−25, −8	−8, −5	+9, –
v0_2, Hornbeck–Molnar band's top, mean	+87	+71	+39	+29
against Kruithof, Heylen	+31, +148	+55, +81	+37, +41	+29, –

Table 6: The ionization coefficient's error (percent, the model less the measurement over the measurement, averaged over each range's points) against Kruithof's and Heylen's measurements [36, 37], v0_1 on Biagi's four groups and v0_2 on his levels, with direct ionization alone and with every excitation of the levels at or above Kr_2^+'s energy ionizing associatively.

Why v0_2 falls short below 50 Td.

- **Not the solver.** LoKI-MC, a Monte Carlo solver with no two-term approximation [3], run on v0_2's own cross sections, gives an ionization coefficient that v0_2's two-term solution matches to +3.7, −4.0, −3.5, −2.5 % at 20, 30, 40 and 50 Td. These differences are within the Monte Carlo's counting error (19 % at 20 Td, 1 % at 50 Td) and a few percent of the two-term approximation. The mean energies agree to 0.15 %. Bordage et al. found the two-term coefficient accurate in krypton against Monte Carlo too [2].
- **Not chiefly the cross sections.** Bordage et al. compared every published set of krypton cross sections, and every one gives coefficients below the measurements at low E/N. They call it a general trend, and ascribe it to the measurements' contribution from Penning ionization of impurities, which

grows as E/N falls [2]. Resolving Biagi's four groups into his 51 levels moves the mean below 50 Td from -54 to -47 %.

- **The measurements disagree.** Heylen's values fall up to $\times 2.2$ below Kruithof's near 30 Td, and Jacques et al. also found values below Kruithof's at the low end of their range in krypton [5].
- **One process of pure krypton.** An atom excited at or above 12.85 eV, the energy of Kr_2^+ relative to two ground-state atoms [32], can ionize on a ground-state atom. Hornbeck and Molnar found Kr_2^+ made by electrons $0.7^{+0.7}_{-0.3}$ V below the energy that makes Kr^+, and traced it to that collision [4]. The electrons' distribution falls by orders of magnitude between those levels and the 13.9996-eV threshold, so at its top the band raises the coefficient $\times 16.8$ at 14 Td, $\times 3.8$ at 30 Td and $\times 1.7$ at 100 Td. At its top 18 of the 21 points below 50 Td lie between the band's two ends, and the 3 of Kruithof's at and below 16.95 Td lie above even its top, which reaches 0.46 to 0.82 of them. The same top makes the model's mean error from 100 to 300 Td +39 %, where direct ionization alone is within 7.6 % rms (Table 6). So a share near 1 cannot hold at every field. It need not: the share is how fast the excited atom meets a partner against how fast it radiates or is quenched, so it grows with the pressure; and a Townsend experiment needs a larger pressure times gap where the coefficient is small, so its low-field points come from higher pressures. The pressures of these measurements are not in Dutton's tables, and no measurement in hand gives the share for krypton, so the band cannot be scored here. It stays at 0 by default, as every unmeasured row does.

What this means for the reference point. The electrons sit at 1.32 to 15.5 Td, where the low ionization coefficient would apply. There direct ionization makes 0.012 % of the ions and Penning ionization the rest, so this error does not reach the answer: the Hornbeck–Molnar band at its top moves the 147-nm share by -0.02 %. It would matter where direct ionization does. No swarm measurement scores the excitation rates, which do decide the answer.

5 Under an electron beam

Three measurements bear on the chain from excitation to the 147-nm photon. They were made under electron beams, so for them v0_2 runs under a beam drive. The drive is a fixture of this note, not part of the model. It deposits the given power as ion pairs, Kr^*, Kr^{**} and secondary electrons in the proportions of a fast electron's degradation in pure krypton. That degradation is the Fowler–Spencer–Fano recursion on the same Biagi cross sections, level by level, from 528 eV: an excitation continues the electron at $E - I$, and an ionization shares $E - I$ equally between the two electrons. Per ion pair it gives:

- W = 27.6 eV;
- 0.488 Kr^{**}, from the levels above the 5s manifold;
- 0.305 Kr^*, from the four 5s levels;
- 0.67 eV of heat, the levels' excess over the energy of the species they make;
- 4.37 eV of sub-excitation secondaries, which the kinetic equation injects with a flat spectrum up to 9.9152 eV.

That is one excitation or ion per 15.38 eV.

The partition. Saito et al. [38] measured scintillation photons and electrons together under 5.49-MeV alpha particles in 1 to 10 atm of krypton. They found 0.55 excited atoms per ion pair, and W_s = 15.4 eV, the energy per photon when every pair recombines. They also confirmed that each recombination gives one photon. The drive's 15.38 eV per excitation or ion is -0.1 % from their W_s. Its excitations per pair (0.79) differ from theirs by +44 %, and its W differs by +16 % from the 23.9 eV their two numbers imply. The two errors cancel in the quantity the light depends on. One photon of 8.43 eV per 15.4 eV is an efficiency of 0.547. v0_2's ceiling, one photon per excitation and per ion of the drive, is 0.548.

Low pump. Morozov et al. [39] measured the power a continuous 8 to 15-keV beam deposits in the gas against the VUV power emitted into 4π. For krypton they found 0.42 ± 0.05, flat from 500 to 1400 hPa and over a tenfold range of beam power. At 1000 hPa, 30 and 300 W cm^{-3} for 1 μs with the light counted

for 50 μs after, v0_2 gives 0.548 and 0.550 (Table 7). The error is +31 %, 2.6 of the measurement's standard errors. At this pump nothing competes with the light: 0.001 Penning ionizations per pair, every ion recombined (1.001 per pair), and every Kr_2^* radiating. So v0_2 sits at its ceiling, which Saito et al.'s measurement supports. Morozov et al. compare their value with calculated maxima of 0.46 and 0.52 for krypton. The 23 % between their value and Saito et al.'s one photon per excitation and per ion (0.547) is not explained in v0_2.

High pump. Eckstrom et al. [40] report the fluorescence efficiency of Kr_2^* near their theoretical limit of about 46 %. They found it under a low-current, long-pulse beam at modest pressures, for pump rates up to 3×10^5 W cm^{-3}. The abstract does not give the pressure of each point. At that pump rate (1 μs, the light counted for 20 μs) v0_2 gives:

- 0.113 at 0.5 atm, 0.345 at 1 atm and 0.529 at 2 atm, that is 79, 37 and 4 % below its ceiling of 0.548;
- at 1 atm, Penning ionization of Kr* by Kr* costs 0.669 photons per pair, because the excimer forms too slowly to outrun it;
- at 1 atm with Kr* + Kr* at its band's low end, 0.451 (18 % below the ceiling), and with Kr* + 2Kr at its band's top, 0.509 (7 %).

Their limit assumes other numbers than v0_2's ceiling, so what the report tests is the loss: little at high pump. v0_2 loses little at 2 atm. If their pressures were near 1 atm, the report leans toward a weaker Penning loss or a faster formation than the defaults. Those are the two inputs that also lead the field-driven uncertainty.

run	measured	kinetic	two-temperature	Penning per pair	photons per pair	T_e (eV)	T_g end (K)
30 W cm^{-3}, 1000 hPa	0.42 $\pm$ 0.05	0.548	0.556	0.000	1.792	0.75	300
300 W cm^{-3}, 1000 hPa	0.42 $\pm$ 0.05	0.550	0.559	0.001	1.798	0.83	300
3×10^5 W cm^{-3}, 0.5 atm	near 0.46	0.113	0.113	1.426	0.369	0.36	1340
3×10^5 W cm^{-3}, 1 atm	near 0.46	0.345	0.345	0.669	1.130	0.45	686
3×10^5 W cm^{-3}, 2 atm	near 0.46	0.529	0.525	0.077	1.730	0.65	439
1 atm, Kr* + Kr* 1.5×10^{-10}		0.451	0.450	0.322	1.476	0.42	624
1 atm, Kr* + 2Kr 5.5×10^{-32}		0.509	0.507	0.136	1.666	0.53	589

Table 7: v0_2 under the beam drive: the VUV efficiency (147-nm energy over the deposited energy) on both closures, and on the kinetic EEDF the Penning ionizations and 147-nm photons per ion pair, the pulse's mean T_e and the final gas temperature. v0_2's ceiling, one photon per excitation and per ion, is 1.793 photons per pair, an efficiency of 0.548; Saito et al.'s ideal is 0.547.

6 The energy ladder

Figure 2 draws v0_2's levels at their formation energies and the rows that move energy between them.

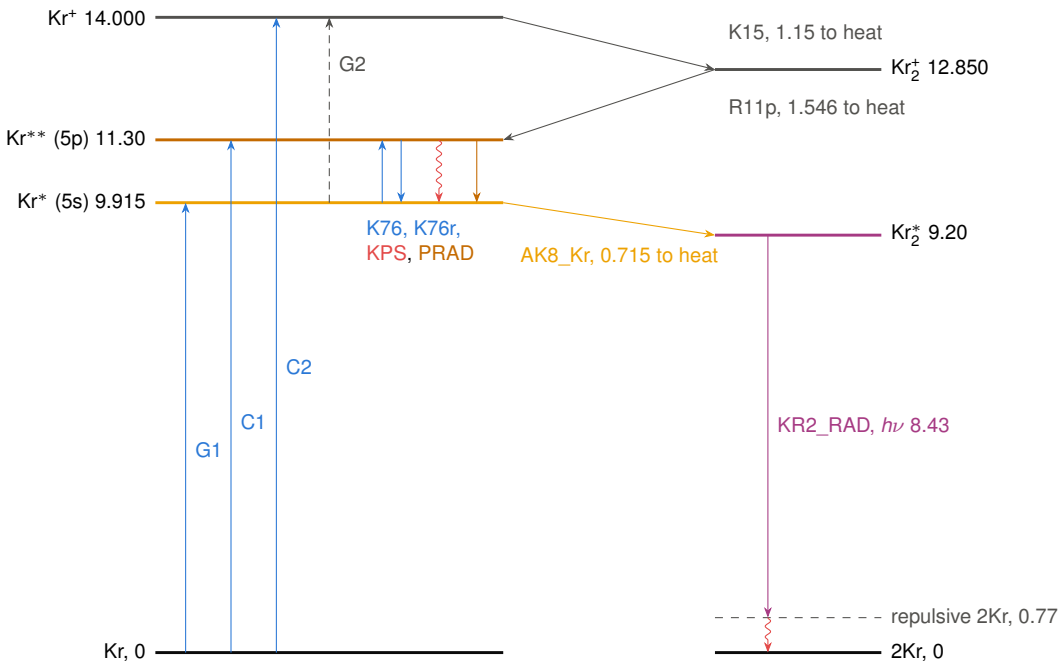


Figure 2: v0_2's energy ladder in eV above ground-state krypton at rest. A bar is a species at its formation energy. Blue arrows are electron impact, which takes its energy from the electron gas; K76r, the 5p $\rightarrow$ 5s arrow beside K76, returns it. The dashed G2 is $Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$, which gives 5.831 eV to the electron. A red squiggle is heat: KPS relaxes the 5p rung on krypton, and PRAD radiates the 1.388-eV gap in the near infrared. The excimer radiates onto the repulsive ground state, which falls apart and heats the gas. G1's four 5s levels, 9.915 to 10.644 eV, make Kr^*; C1's levels, 11.30 to 13.60 eV, cascade to the 5p rung; each heats the gas with its excess. HM, the Hornbeck–Molnar ionization of the levels above Kr_2^+'s energy, is a band at 0 and not drawn.

7 Where v0_2 is not yet reliable

- **Kr^* + 2Kr.** The excimer's formation rate is the largest term of the propagated uncertainty: its measurements spread by a factor of 3.2. Its temperature law is measured to 451 K, and the reference point ends at 807 K.
- **The field-driven regime is unscored.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field. Its cross sections are measured to $\pm$35 to 40 %. Those of the two $J = 1$ levels are not measured; they are the band's low end.
- **The rows with no measurement.** Kr_2^* + Kr_2^* and the stepwise ionizations of Kr^* and Kr^{**} are 0 by default. Each at its band's top cuts the 147-nm share by a fifth to a half (Table 2).
- **Kr^* + Kr^*.** Its 5.0×10^{-10} cm^3s^{-1} is a 1974 measurement [18] as Kannari et al. print it [11]. The e-beam report at high pump leans toward a lower value (Section 5).
- **The e-beam efficiency at low pump** is off Morozov et al.'s by +31 %, and v0_2 holds no process that would explain the difference.
- **The ionization coefficient below 50** Td has a mean error of -47 %, as every published cross-section set falls below these measurements [2]. This does not reach the reference point, where Penning ionization makes the ions.
- **The Hornbeck–Molnar share** of the levels above 12.85 eV is not measured for krypton and is 0 by default. At its top it brackets most of the low-field measurements, overshoots those from 100 to 300 Td (+39 %) and moves the reference point by -0.02 %.
- **Processes v0_2 does not carry**, each needing a measured rate before it can enter:
  - Kr_3^+ and its recombination;
  - electron collisions with Kr_2^*: ionization, quenching and the mixing of its two states (v0_2 carries

one state at the slow component's 270 ns);
  - Penning ionization between Kr^* and Kr_2^*, and of Kr^{**};
  - the redissociation of Kr_2^+ and Kr_2^* in hot gas;
  - the resonance lines of the two $J = 1$ 5s levels.

  Three-body recombination $Kr^+ + e^- + e^-$ [33] is not carried either: at 1 atm Kr^+ converts to Kr_2^+ in 7 ns.
- **The kinetic EEDF's steady shape.** Between renewals the EEDF keeps its steady shape, which the first nanosecond and the afterglow strain. The two-temperature closure reads a tail table made for LIGHT's Kr/F_2 network on Biagi's four groups (its $F_2 = 0$ column); the default closure does not use it.

How the numbers were made. Every number in this note is computed from the model when the note is built. None is typed by hand except the measurements, which carry their sources. The runs are cached under a hash of the model's code, so a change to the model reruns them all (code `cfc67a01cba9`). The kinetic runs take about 7 minutes each; the two-temperature runs take seconds.

A Reaction rates

The rows of v0_2 with the values they run with and their sources. Electron-impact rows use their cross sections over the kinetic EEDF (or the two-temperature closure). A band gives its default and its two ends.

row	reaction	value	source
G1	$e^- + Kr \rightarrow e^- + Kr^*(5s)$	the four 5s levels, 9.9152 to 10.644 eV, each at its threshold; the excess over 9.9152 to heat	Biagi, Magboltz 8.97 [1, 6, 7]
C1	$e^- + Kr \rightarrow e^- + Kr^{**}$	the 5p, 4d, 6s and higher levels, 11.30 to 13.60 eV, each at its threshold; the excess over 11.30 to heat	Biagi, Magboltz 8.97
HM	$e^- + Kr\ (+ Kr) \rightarrow Kr_2^+ + 2e^-$	the share of the excitations of the 20 levels at or above 12.85 eV; band 0 (default) to 1	no measurement for krypton; the process, Hornbeck and Molnar [4]
C2	$e^- + Kr \rightarrow Kr^+ + 2e^-$	13.9996 eV	Biagi, Magboltz 8.97
G6	$e^- + Kr$, elastic	the momentum transfer cross section	Biagi, Magboltz 8.97
K76	$e^- + Kr^* \rightarrow e^- + Kr^{**}$	the lumped measured cross section, 1.1×10^{-14} cm^2 at 6 eV; band $\times$ 0.5 to 1.4	Jung et al. [10, 9]; Kannari's fit as a comparison [11, 12]
K76r	$e^- + Kr^{**} \rightarrow e^- + Kr^*$	detailed balance, g 12/36 over the 1.388-eV gap	
KPS	$Kr^{**} + Kr \rightarrow Kr^* + Kr$	1.7×10^{-10} cm^{3}s^{-1}; band 3.0×10^{-11} to the default	Gazeli et al. [14], one level, products unresolved; the low end lies in the range Chang and Setser measured for argon's 4p levels in argon, 2×10^{-11} to 10^{-10} [16]
PRAD	$Kr^{**} \rightarrow Kr^* + h\nu$, 760–830 nm	τ = 29.0 ns, the $2J + 1$ mean of the ten 5p levels	Dzierżęga et al. [17]; a TALIF lifetime of one level, 34.1 ns, as a cross-check [15]
G2	$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}; band 1.5×10^{-10} to the default	Leichner and Ericson [18] as Kannari et al. print it [11]; the low end has no published measurement
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.64}$; band to 5.5×10^{-32}	Ueda et al. at 0.5–1 atm [19]; afterglows [20, 21, 22] and Hughes's value [23] inside the band; the law, Wieme and Lenaerts [24]
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$, 147 nm	τ = 270 ns, $h\nu$ = 8.43 eV	Morikawa et al., 270 $\pm$ 5 ns [25]; 250 $\pm$ 20 [20], 280 $\pm$ 30 [23]
PEN_KR2	$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0; band to 5.0×10^{-10}	no measurement; Kannari et al.'s value is the top [11]
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.4×10^{-31} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.75}$	the mean of three measurements [26, 27, 28]; the law, Pérez-Ríos and Greene [29]
R11p	$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	1.6×10^{-6} (300 K$/T_e$)$^{0.55}$ cm^{3}s^{-1}, $\pm$10 %	Shiu and Biondi [30]; Mikus within 8 % [31]
KSTEP_S	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0; band to $1.5 \times 10^{-7}\ T_e^{0.68} e^{-4.04/T_e}$ cm^{3}s^{-1}	no measurement; Vriens' calculation [13] as Kannari et al. print it [11]
KSTEP_P	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0; band to $2.0 \times 10^{-7}\ T_e^{0.62} e^{-2.53/T_e}$ cm^{3}s^{-1}	the same

Table 8: v0_2's rows. Rates in cm^3 s^{-1} or cm^6 s^{-1}, T_e in eV. Two rows sit in the optional group `electron`, off by default: a stepwise-ionization template of Kr^* and the superelastic return of the 5s levels by detailed balance. Kr_2^+ lies 1.15 eV below Kr^+ + Kr [32]; Kr_2^*'s 0.715-eV well is an estimate that moves energy only between storage and heat.

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